

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department of Mechanical and Production Engineering



Project Report

Course Code: IPE 3217

Course Name: Operations Research II

Project Title: Queuing system analysis of the ablution corner of AUST Prayer Room

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PREAMBLE

Queueing theory provides mathematical techniques for analyzing waiting lines and service systems. It can be used to evaluate the relationship between customer arrivals, service capacity and waiting time, and to determine an appropriate service capacity for a given system.

In this project, the ablution facility of the prayer room at AUST was selected as a real-life queueing system.. The individual water taps were considered as servers, while students arriving for wudu were considered customers.

A thorough study on the system was done during peak periods and data on customer arrivals, service times and queue conditions were collected. The collected data were then used to estimate the arrival and service rates and select an appropriate queueing model. Finally, the current system was analyzed and alternative service capacities were compared to determine a suitable way of reducing waiting while maintaining an appropriate service capacity.

PROBLEM IDENTIFICATION

During peak periods, it was observed that multiple students arrived within a short period to perform wudu. This resulted in the formation of a queue.

OBSERVATIONS AND DATA COLLECTION

Study Location: AUST Prayer Room Ablution Corner

Service Time Data Table:

Person	Recorded Wudu Time	Service Time (sec)
1	56 sec	56
2	45 sec	45
3	1 min 17 sec	77
4	1 min 10 sec	70
5	45 sec	45
6	1 min 21 sec	81
7	2 min 26 sec	146
8	57 sec	57
9	2 min 28 sec	148
10	1 min 10 sec	70
11	1 min 58 sec	118
12	1 min 53 sec	113
13	2 min 07 sec	127
14	2 min 09 sec	129
15	1 min 57 sec	117
16	2 min 05 sec	125
17	2 min 6 sec	126
18	2 min 33 sec	153
19	2 min 37 sec	157
20	3 min 07 sec	187
21	2 min 05 sec	125
22	2 min 13 sec	133
23	1 min 56 sec	116
24	1 min 36 sec	96
25	1 min 46 sec	106

Arrival Time Data Table:

Arrival no.	New arrival time (mm:ss)	Interarrival time (seconds)
1	00:12.99	00:12.99
2	00:14.98	00:01.99
3	00:56.75	00:41.77
4	00:59.90	00:03.15
5	01:22.17	00:22.27
6	02:05.99	00:43.82
7	02:19.77	00:13.78
8	02:32.67	00:12.90
9	02:45.12	00:12.45
10	03:15.29	00:30.17
11	03:58.43	00:43.14
12	04:10.59	00:12.16
13	04:14.35	00:03.76
14	04:50.94	00:36.59
15	05:08.47	00:17.53
16	05:55.62	00:47.15
17	06:31.56	00:35.94
18	06:36.17	00:04.61
19	06:39.06	00:02.89
20	07:30.98	00:51.92
21	08:03.01	00:32.03
22	08:04.11	00:01.10
23	08:04.88	00:00.77
24	09:07.27	01:02.39
25	10:08.70	01:01.43

ANALYSIS

Performing wudu takes a minimum amount of time. The lognormal distribution reflects human behaviour quite perfectly, which shows a realistic baseline for the time taken to perform wudu and it gets a right skewed tail as some individuals take longer.

The selected model should be M/LN/S:GD/ ∞ / ∞

CALCULATION

For arrival:

$$\text{Average Interarrival Time} = \frac{608.70}{25} = 24.35 \text{ seconds}$$

$$\text{Average arrival rate } (\lambda) = \frac{25}{608.70/60} = 2.46 \text{ arrivals/minute}$$

$$\text{Arrival rate } (\lambda) \text{ per hour} = 2.4643 \times 60 = 147.86 \text{ arrivals/hour}$$

For service:

$$\text{Average service time} = 108.92 \text{ sec}$$

$$\text{Average service rate } (\mu) = \frac{1}{108.92/60} = 0.5513 \text{ customers/minute per server}$$

$$\text{Service rate } (\mu) \text{ per hour} = 0.5513 \times 60 = 33.08 \text{ customers/hour per server}$$

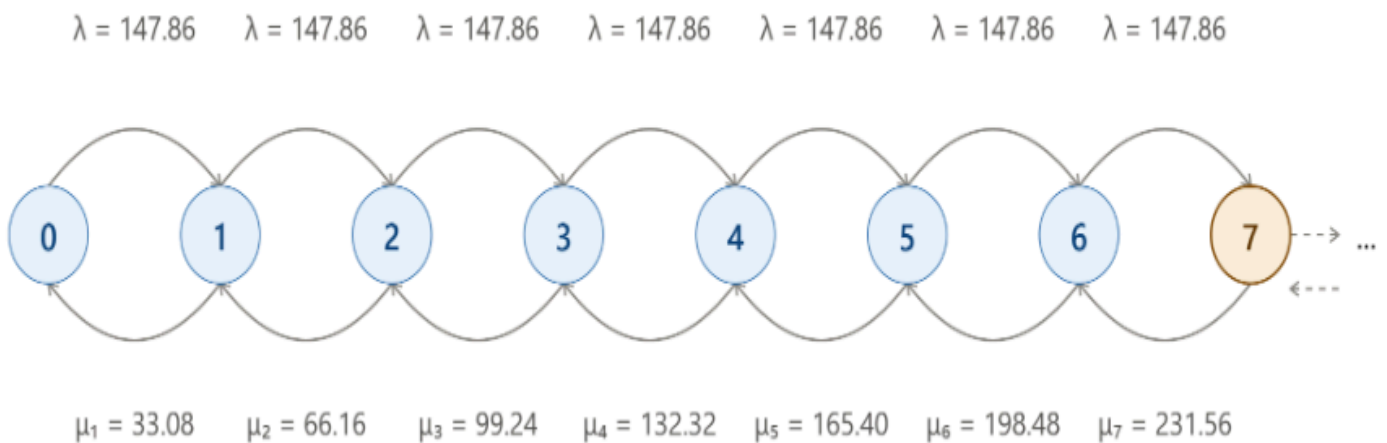
$$\text{Number of servers (S)} = 7 \text{ faucets}$$

$$\text{Total service capacity} = 7 \times 33.08 = 231.56 \text{ customers/hour per server}$$

RESULT AND DISCUSSION

From the collected data, the average arrival rate was 147.86 students per hour and the average service rate of each faucet was 33.08 students per hour. As a total of 7 faucets were there, the total service capacity was approximately 231.56 students per hour. Even though the total service capacity was higher than the average arrival rate, temporary queues were still observed, especially during the main congregational prayer as students rushed to do wudu to attend the prayer that had already started. Because of this a large number of students arrived within a very short time resulting in the formation of the temporary queue.

Rate Diagram:



CONCLUSION

In this project, the queueing system of the AUST Prayer Room ablution corner was analyzed. The students performing wudu were considered as customers while the water faucets were considered as servers. During the peak periods, the arrival times, service times and queue length were observed and recorded. It was seen that there was a formation of a temporary queue due to the sudden rapid increase in the arrival rate.

Based on our observations and the recorded data several possible improvements were identified:

- The side benches can be removed or rearranged so that more space can be utilized for installing additional faucets where possible.
- Instead of everyone trying to perform wudu and attend only the main congregation, students can be encouraged to make use of the consecutive congregations that are held before and after the main one.
- A separate entrance and exit can be provided to improve the movement of students. If separate entrance and exit points cannot be arranged, the common entrance and exit area can be made wider which would facilitate movement and reduce congestion.

At the end we can say that the formation of the queue at the ablution corner is not really due to the lack of average service capacity but due to sudden rush during the prayer time. As it naturally takes time to perform ablution which might eventually form a queue during a rush. So to reduce the waiting time as well as the queue length, properly utilizing available space, installing additional faucets where feasible, encouraging the students to spread out across the existing congregations are viable options.